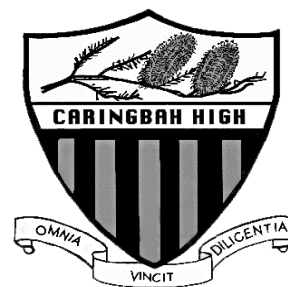


Caringbah High School
Year 11 Mathematics 2014
Semester 2 Examination



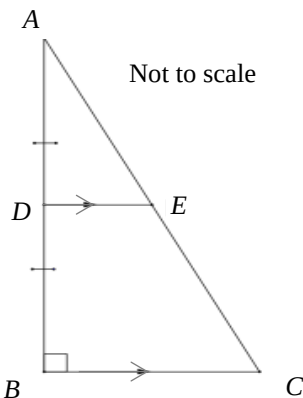
Time Allowed: 2 hour + 5 minutes reading

Instructions

- Start each question in a new booklet.
- Answers unsupported by working may not attract full marks.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Attempts that show an understanding of concepts may attract some marks.

Question 1 (13 marks)

Marks

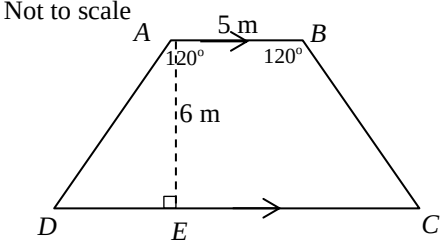
- a) Find the midpoint of the interval joining the points $A(5, -2)$ and $B(-3, -6)$. 1
- b) State the gradient of the line $3y - 2x + 5 = 0$. 1
- c) Sketch the graph of $y = 4 - x^2$, showing all intercepts on axes. 2
- d)  The $\triangle ABC$ is right-angled at B .
 $BC \parallel DE$.
 $AD = DB$.
 i) Show that $\triangle ADE$ is right-angled. 1
 ii) Prove that $\triangle ABC \sim \triangle ADE$. 2
- e) Sketch the graph of $y = 3^x$. 1
- f) State the range of the function $xy = 4$. 1
- g) If $f(x) = x^2 + 4x - 1$, find all possible values for x if $f(x) = 20$. 2
- h) Find the derivative of $3x^2 - 2x + 1$. 2

Question 2 (13 marks)

- a) The points $A(-3, 2)$, $B(5, 1)$, $C(7, -2)$ and D form a parallelogram.
- i) Plot these points on a number plane and show the coordinates of D . 2
- ii) Find the coordinates of the point E where AC intersects with BD . 1
- b) Determine the natural domains of the following functions.
- i) $y = \frac{x}{x+1}$ 1
- ii) $y = \sqrt{x-4}$ 1
- c) Find the distance between the points $P(5, 9)$ and $Q(-3, 4)$ correct to 3 significant figures. 2
- d) Find the derivatives of the following functions
- i) $y = 4x^{\frac{3}{5}}$ 2
- ii) $y = \frac{5}{2x^4}$ 2
- iii) $y = \sqrt[3]{4x-2}$ 2

Question 3 (13 marks)

Marks

- a)  $ABCD$ is an isosceles trapezium with $AB \parallel DC$ and $\angle DAB = \angle CBA = 120^\circ$. The perpendicular height AE is 6 m. $AB = 5$ m.
- Show that $DE = 2\sqrt{3}$ m. 2
 - Find the area of the trapezium. 2

- b) Sketch the graph of these parabolas and find the coordinates of their vertex.

i) $y = (x - 2)(x + 4)$ 2

ii) $y = 3x(6 - x)$ 2

- c) Find the derivatives of these functions.

i) $f(x) = (3x^2 + 5)^4$ 2

ii) $f(x) = (2x^3 - 7)^3(4x - 1)$, in simplest factorised form. 3

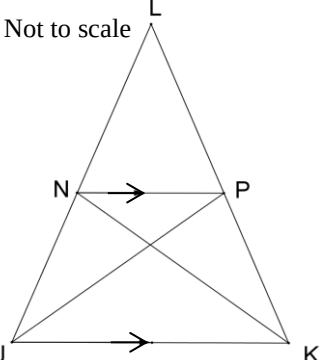
Question 4 (13 marks)

- a) Find the equation of the line that passes through the points $A(5, -2)$ and $B(2, 1)$. 3

- b) Find the point of intersection of the straight lines $x + 2y = 5$ and $x - y = 2$. 2

c) i) Find $\lim_{x \rightarrow -2} \frac{x^2 - 25}{x - 5}$ 1

ii) Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$ 1

- d)  In the diagram, JKL is an isosceles triangle where $JL = KL$ and JK is parallel to NP .

- i) Show that triangle LNP is an isosceles triangle. 2

- ii) Show that $JN = KP$. 1

- iii) Show that $\triangle NJK \equiv \triangle PKJ$. 3

Question 5 (12 marks)**Marks**

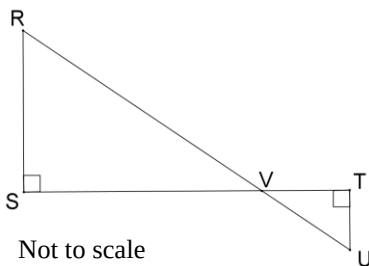
- a) State, with reasons, whether the function $y = \sin x$ is an odd or an even function. 1
- b) Find the perpendicular distance from the point $(2, 5)$ to the line $3x + 4y = 1$. 2
- c) Find the equation of the line through the point $(-2, 6)$ that is parallel to the line $2x - 4y + 3 = 0$. 2
- d) Find the equation of the tangent to the curve $y = x^3 - 2x^2 + 3x - 1$ at the point $(1, 1)$. 3
- e) Show that the line $3x - 2y + 14 = 0$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 13$. 2
- f) Shade the region defined by $y \leq x^3 + 1$ 2

Question 6 (12 marks)

- a) Find the sum of the internal angles of a decagon. 1
- b) The exterior angle of a regular polygon is 24° . 2
How many sides does the polygon have?
- c) i) On the same set of axes, sketch the graphs of $y = |2x - 1|$ and $y = 3$. 2
ii) Find the points of intersection of these two graphs. 2
- d) Differentiate from first principles $y = x^2 - 3x + 1$. 2
- e) Differentiate $y = \frac{(3x - 2)^3}{5x + 1}$ 3

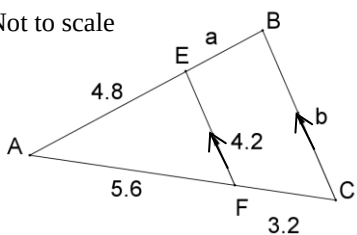
Question 7 (12 marks)

Marks

- a)  2
- Rudi used similar triangles to measure the width of a creek. The field diagram shown shows two points, R and S, on opposite banks of the creek. Using line of sight, he constructed a smaller similar triangle VTU on the land.
- He measured the following distances.
SV = 25 m, VT 5 m, TU = 12 m and VU = 13 m.
Find the width of the creek RS.
- b) Sketch the graph of $y = \sqrt{25 - x^2}$ 2
- c) Find the equation of the line that passes through the origin and the point of intersection of the lines $2x + 3y - 9 = 0$ and $x - y - 2 = 0$. 4
- d) A function is defined as $f(x) = \begin{cases} x^3 & x < 0 \\ x + 1 & x \geq 0 \end{cases}$
- i) Explain why $f(x)$ is not a continuous function. 1
- ii) Evaluate $f(-2) + f(3)$. 1
- iii) Sketch the graph of $y = f(x)$. 2

Question 8 (12 marks)

Marks

- a) i) Differentiate $y = \sqrt{x+3}$. 2
- ii) Find the gradient of the curve $y = \sqrt{x+3}$ at the point (1, 2). 1
- iii) Find the equation of the normal to the curve $y = \sqrt{x+3}$ at the point (1, 2). 2
- b) Not to scale  In this diagram $FE \parallel CB$. 2
- $AE = 4.8, AF = 5.6, FE = 4.2$ and $FC = 3.2$.
- Find values for a and b .
- c) Sketch the graph of $y = \frac{-1}{x+2}$, showing all asymptotes. 2
- d) The normal to the curve $y = ax^3 - 1$ at $x = 1$ meets the positive x -axis at an angle of 120° . Find the value of a . 3

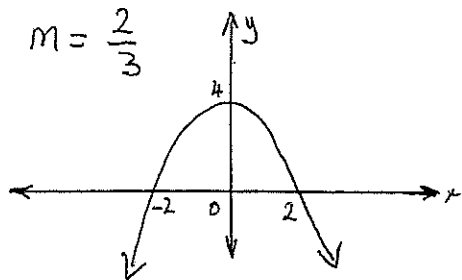
END OF PAPER

Question 1

a) $(1, -4)$

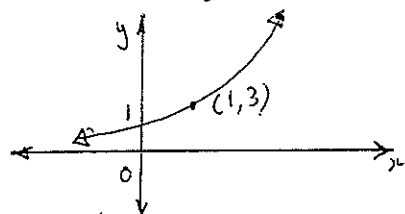
b) $m = \frac{2}{3}$

c)



d) $\angle ADE = \angle ABC$ corresponding angles, $DE \parallel BC$
 $\therefore 90^\circ$ - given $\therefore \angle ADE$ is right-angled

e)



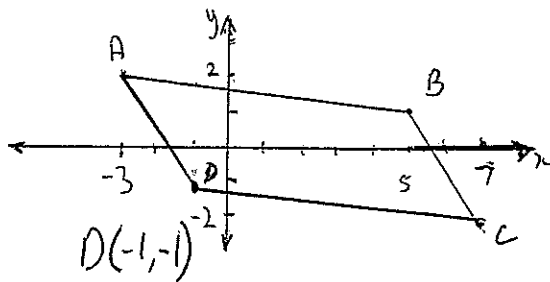
f) All real y , $y \neq 0$

g) $x^2 + 4x - 1 = 20$
 $x^2 + 4x - 21 = 0$
 $(x+7)(x-3) = 0$
 $x = -7$ or $x = 3$

h) $6x - 2$

Question 2

a) i)



ii) $E(2, 0)$

b) i) D: All real x , $x \neq -1$

ii) D: $x \geq 4$

c) $d = \sqrt{(5+3)^2 + (9-4)^2}$
 $= \sqrt{89}$
 $= 9.43$

d) i) $y' = \frac{12}{5} x^{-\frac{2}{5}}$

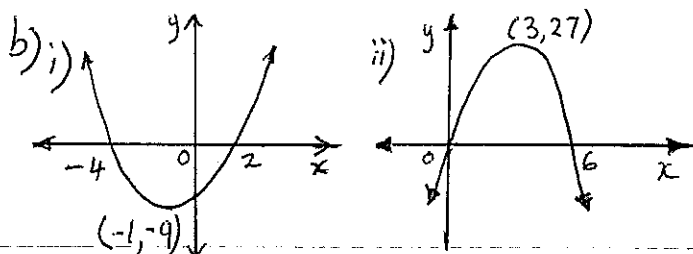
ii) $y = \frac{5}{2} x^{-4}$ $y' = -10x^{-5}$
 $= -\frac{10}{x^5}$

iii) $y = (4x-2)^{\frac{1}{3}}$
 $y' = \frac{1}{3} (4x-2)^{-\frac{2}{3}}$
 $= \frac{1}{3 \sqrt[3]{(4x-2)^2}}$

Question 3

a) $\angle DAE = 30^\circ$ $\tan 30^\circ = \frac{DE}{6}$
 $\frac{1}{\sqrt{3}} = \frac{DE}{6}$
 $DE = \frac{6}{\sqrt{3}}$
 $= 2\sqrt{3}$

ii) $A = \frac{1}{2}(a+b)$
 $= \frac{6}{2}(5 + (5+4\sqrt{3})) m^2$
 $= 30 + 12\sqrt{3} m^2$



c) i) $f'(x) = 24x(3x^2+5)^3$

ii) $u = (2x^3-7)^3$ $v = (4x-1)$
 $u' = 18x^2(2x^3-7)^2$ $v' = 4$

$f'(x) = 18x^2(2x^3-7)^2(4x-1) + 4(2x^3-7)^3$
 $= 2(2x^3-7)^2 [9x^2(4x-1) + 2(2x^3-7)]$
 $= 2(2x^3-7)^2 (40x^3 - 9x^2 - 14)$

Question 4

a) $m = \frac{3}{-3}$ Eqn $(y-1) = -1(x-2)$
 $= -1$ $y-1 = -x+2$
 $x+y-3=0$

b) $x+2y=5$ ①
 $x-y=2$ ②
 $① - ② \cdot 3y = 3$
 $y = 1$ $x = 3$

Q4c

$$i) \frac{4-25}{-2-5} = 3$$

$$ii) \lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{(x-5)} = 10$$

d) $\angle LNP = \angle LJK$ corres. $\angle$ s, $NP \parallel JK$.
 $\angle LPN = \angle LKJ$ " " , $NP \parallel JK$.
 $\angle LJK = \angle LKJ$ angles opp. equal sides of an isos. Δ .
 $\therefore \angle LNP = \angle LPN$
 $\therefore \Delta LNP$ is isosceles

ii) $JL = KL$ given
 $LN = LP$ equal sides of isos. Δ ①

$$NJ = JL - LN$$

$$PK = KL - LP \quad JL - LN = KL - LP \quad \text{from ①}$$

$$\therefore NJ = PK.$$

iii) In ΔNJK and ΔPKJ

S JK is common

A $\angle NJK = \angle PKJ$ angles opp. equal sides of isos ΔLJK .

S

$$NJ = PK \quad \text{from ii)}$$

$$\therefore \Delta NJK \equiv \Delta PKJ \quad \text{SAS}$$

Question 5

a) $\sin(-\theta) = -\sin \theta$ so $\sin x$ is odd
 NB. You must reference the question. ie use $\sin x$.

$$b) d = \left| \frac{(3 \times 2) + (4 \times 5) - 1}{\sqrt{3^2 + 4^2}} \right|$$

$$= 5$$

$$c) m = \frac{1}{2} \quad (y-6) = \frac{1}{2}(x+2)$$

$$2y-12 = x+2$$

$$x-2y+14=0$$

$$d) y' = 3x^2 - 4x + 3 \quad \text{Eqn of tangent}$$

$$x=1 \quad m=2$$

$$(y-1) = 2(x-1)$$

$$y-1 = 2x-2$$

$$2x-y-1=0$$

e) Circle centre $(1,2)$ rad = $\sqrt{13}$

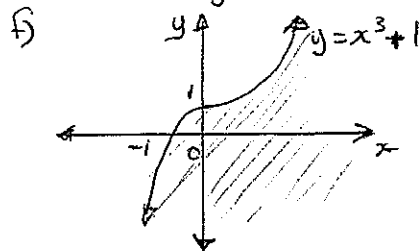
Using $\perp$ dis from centre

$$d = \left| \frac{(3 \times 1) - (2 \times 2) + 14}{\sqrt{3^2 + (-2)^2}} \right|$$

$$= \left| \frac{13}{\sqrt{13}} \right|$$

$$= \sqrt{13} \quad (= \text{radius})$$

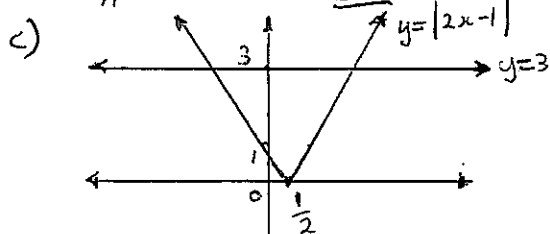
$\therefore 3x-2y+14=0$ is a tangent.



Question 6

$$a) 180^\circ \times 8 = 1440^\circ$$

$$b) \frac{360}{n} = 24 \quad \therefore n = 15$$



$$ii) 2x-1=3 \quad \text{or} \quad 2x-1=-3$$

$$x=2$$

$$y=3$$

$$x=-1$$

$$y=3$$

$$d) f(x) = x^2 - 3x + 1$$

$$f(x+h) = (x+h)^2 - 3(x+h) + 1$$

$$= x^2 + 2xh + h^2 - 3x - 3h + 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 - 3x - 3h + 1) - (x^2 - 3x + 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 3)}{h}$$

$$= 2x - 3$$

$$e) y' = \frac{vu' - uv'}{v^2} \quad u = (3x-2)^3 \quad v = (5x+1)$$

$$u' = 9(3x-2)^2 \quad v' = 5$$

$$= \frac{9(3x-2)^2(5x+1) - 5(3x-2)^3}{(5x+1)^2}$$

$$= \frac{(3x-2)^2 [9(5x+1) - 5(3x-2)]}{(5x+1)^2}$$

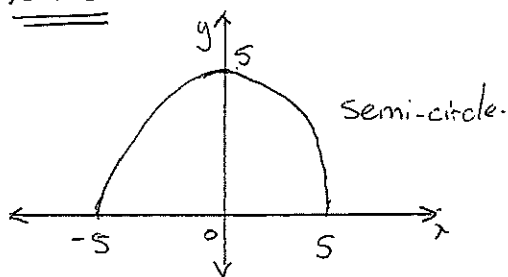
$$= \frac{(3x-2)^2 (5x+19)}{(5x+1)^2}$$

Question 7

a) $\frac{x}{25} = \frac{12}{5}$

$x = 60$

b)



c)

$L_1 + kL_2 = 0$

$(2x + 3y - 9) + k(x - y - 2) = 0$

$x=0, y=0 \Rightarrow -9 - 2k = 0$

$k = -4\frac{1}{2}$

$(2x + 3y - 9) - \frac{9}{2}(x - y - 2) = 0$

$x - 3y = 6$

(Alternatively, solving simultaneously to find point of intersection $(3, 1)$ giving $m = \frac{1}{3}$)

d) i) at $x=0$ $\text{FOV} = 1$
but as $x \rightarrow 0$ from the negative.

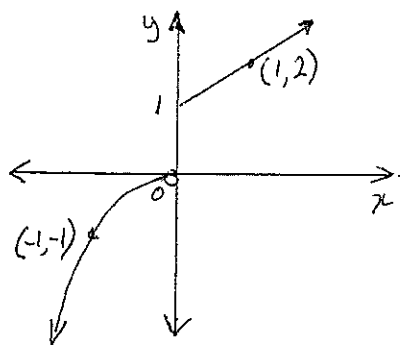
$\text{FOV} \rightarrow 0^3 = 0$

$\therefore$ There is a gap in the graph at $x=0$

ii) $f(-2) + f(3) = -8 + 4 = -4$

NB: You must reference the question on the paper.

iii)



Question 8

a) i) $y = (x+3)^{\frac{1}{2}}$

$y' = \frac{1}{2}(x+3)^{-\frac{1}{2}}$

$= \frac{1}{2\sqrt{x+3}}$

ii) $x=1 \quad m = \frac{1}{4}$

iii) $m_N = -4$ Eqn of normal

$(y-2) = -4(x-1)$

$y-2 = -4x+4$

$4x + y - 6 = 0$

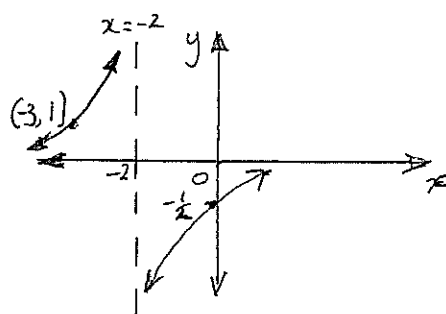
b) $\frac{a}{4 \cdot 8} = \frac{3 \cdot 2}{5 \cdot 6}$ ratio of intercepts on parallel lines.

$a = 2.74$ (2d.p)

$\frac{b}{4 \cdot 2} = \frac{8 \cdot 8}{5 \cdot 6}$ ratio of sides of similar Δ s.

$b = 6.6$

c)



d) $y' = 3ax^2$

at $x=1 \quad m_T = 3a \quad m_N = -\frac{1}{3a}$

$m_N = \tan 120^\circ$

$= -\sqrt{3}$

$-\frac{1}{3a} = -\sqrt{3}$

$a = \frac{1}{3\sqrt{3}} = \left(\frac{\sqrt{3}}{9}\right)$